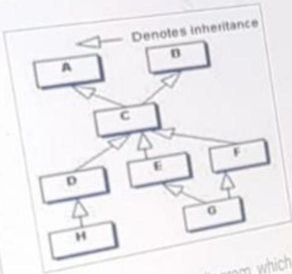
HirePro

Instructions

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

QUESTION: 3 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

Consider the figure given below:

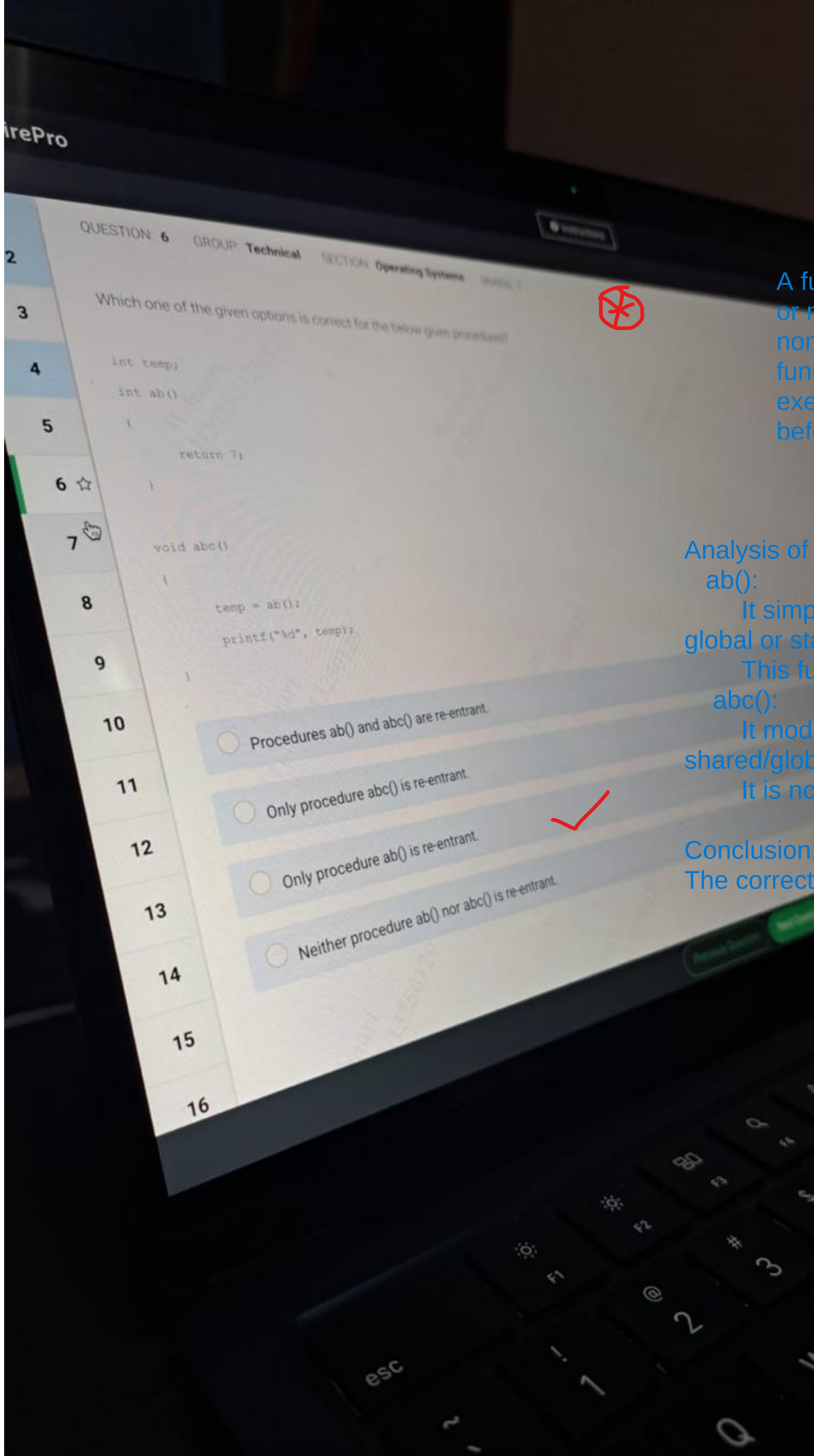



Referring to the above diagram, which one of the statement given in options is true?

- Class C has all non private data members of class A.
- Class B and C have same non-private data members.
- Class A and B have same non-private data members.
- Class A and C have same non-private data members.

Previous Question

Next Question




A function is considered reentrant if it does not rely on or modify shared/global/static data and does not call non-reentrant functions. This means that a reentrant function can be safely interrupted in the middle of its execution and then safely called again ("re-entered") before the previous executions are complete.

Analysis of Reentrancy:

ab():

It simply returns a constant value and does not modify any global or static variable.

This function is reentrant.

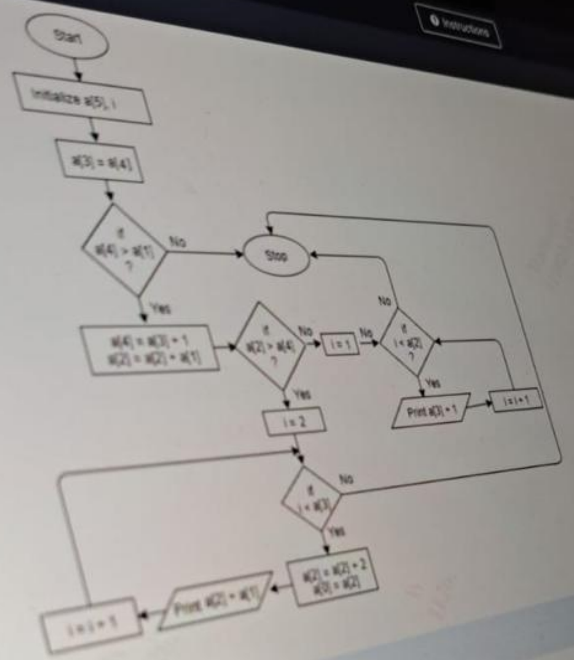
abc():

It modifies the global variable temp and hence relies on shared/global data.

It is not reentrant.

Conclusion:
The correct answer is "Only procedure ab() is re-entrant."

Instructions

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- 6 1 8 2 1
- 5 2 9 1 2
- 4 2 3 4 7
- 5 1 3 4 4

Previous Question

Next Ques

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Algorithm:

```
1. Start
2. Initialize j, m = 3, k
3. Set j = 3
4. Repeat step 5 until j < 6, m step by 1
5. Compute j = m + 2
   End loop
6. Compute m = j - 1
7. Check if j < m
8. then m = j + m
9. else j = j * m
   end if
10.Compute j = m + 2
11.Check if j = 8 then go to step 7
   end loop
12.Compute k = j + m
13.Print k
14.Stop
```

- 12
- 10
- 7
- 11

QUESTION: 2 GROUP: **Technical** SECTION: **Operating Systems** Mark(s): 2

2 ☆

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Which one of the given options is correct for the below given procedures?

```
int temp;
int ab()
{
    return 7;
}

void abc()
{
    temp = ab();
    printf("%d", temp);
}
```

- ○ Only procedure ab() is re-entrant.
- ○ Procedures ab() and abc() are re-entrant.
- ○ Only procedure abc() is re-entrant.
- ○ Neither procedure ab() nor abc() is re-entrant.

Instructions

ESTION: 25 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

onsider the table given below:

STUDENT

| STD_ID | STD_NAME | DATE_OF_BIRTH | PHONE | DEPT_NAME |
|---|---|---|---|---|
| 1 | aa | 10-10-1991 | 9988776655 | maths |
| 2 | bb | 11-09-1990 | 9977776655 | computers |
| 3 | cc | 02-09-1991 | 9977576655 | stats |
| 4 | dd | 02-09-1991 | 9977574655 | maths |

Which one of the query given below will display the names of students in the departments other than Maths and Computer?

- SELECT std_name FROM student WHERE dept_name NOT EXISTS ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT EXISTS IN ('maths','computers');

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Previous Question

TIME LEFT 00:39:18

26 ☆

27

**Output Format:**

N lines of output contain t, number of pages selected for printing for each printer.

**Sample Input 1:**

5
4
5
2
3
6

**Sample Output 1:**

3
2
2
2
4

**Explanation 1:**

The total number of printers available are 5.
For the first printer, T=4. The number of factors of integer '4' are 3 (1,2 and 4), which is printed in the first line of the output.
For the second printer, T=5. The number of factors of integer '5' are 2 (1 and 5), which is printed in the second line of the output.
Similarly, the number of pages that can be printed is found out for the following printers.

**Sample Input 1:**

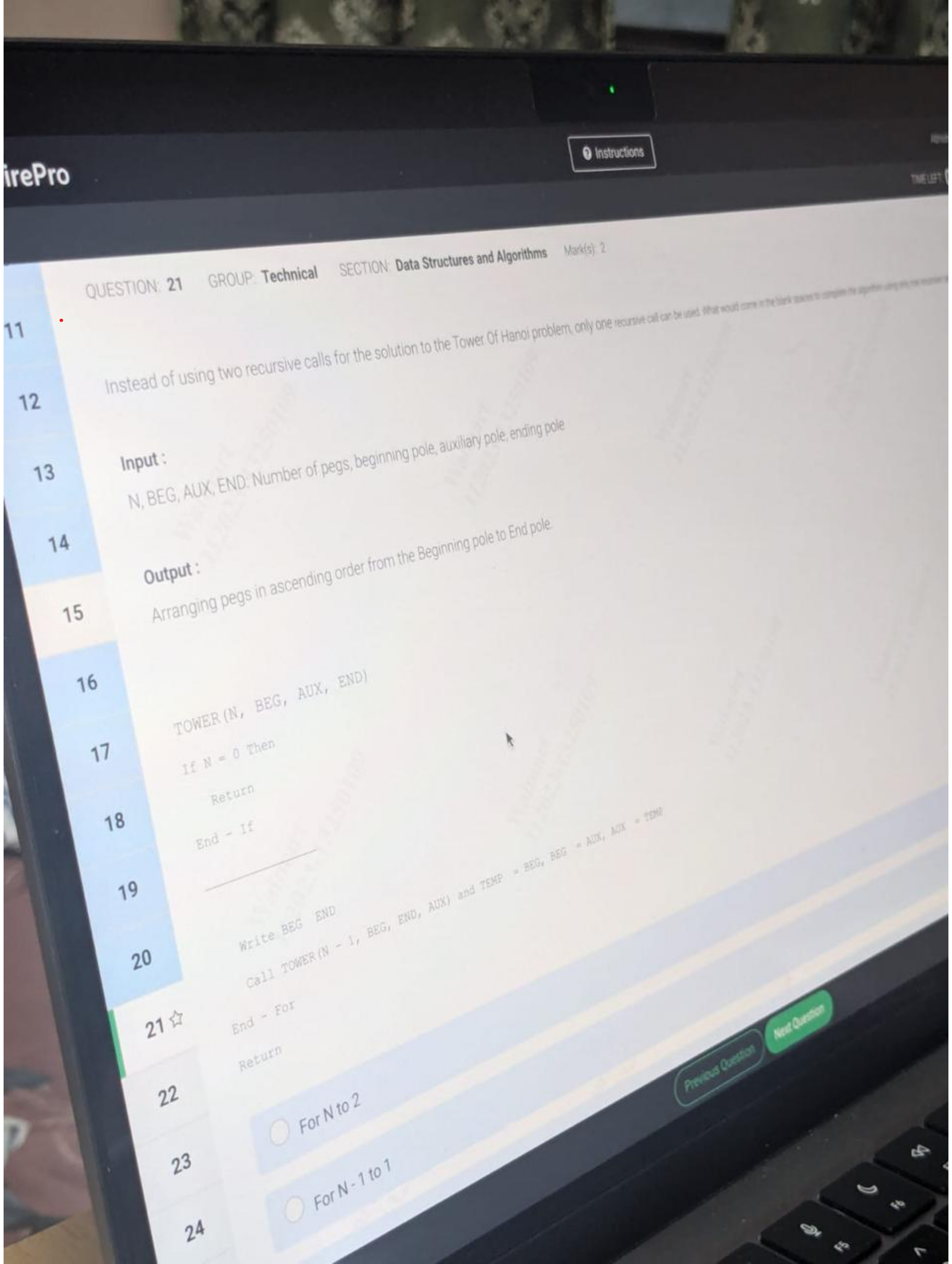
3
12
22
9

**Sample Output 1:**

6
4
3

are 3.
...rs of integer '12' are 6 (1,2,3,4,6 and 12), which is printed in the first line of the output.
...er '22' are 4 (1,2,11 and 22), which is printed in the second line of the output.
...1,3 and 9), which is printed in the third line of the output.

Next Question




For Temp = 1 to N - 1

Pro

Instructions

Abhishek Maurya

QUESTION: 27 GROUP: Coding SECTION: Coding 2 Mark(s): 30

TIME LEFT 00:38:11

A robot Alex is programmed to manipulate human speech for further syntactical processing. Alex's job is to hear the user's voice and capture one sentence for further assessment. A part of the processing for syntactical assessment consists of the steps given below:

I) If a vowel is immediately followed by a consonant in the sentence, they exchange their positions.
II) Next alphabet traversal happens through the exchanged vowel.
III) No exchange happens if a consonant is followed by a vowel or vowel is followed by a vowel or a consonant followed by a consonant.
IV) The special characters [\t + $ - @ ! & * ( ) [ ] # ^] and the spaces between words in the sentence have be eliminated.

Write a program to print the resultant string that the robot Alex gives as output.

Illustrate this scenario by reading the input from STDIN and the output from STDOUT. You should not write arbitrary strings while reading the input and while printing as these contribute to the standard output.

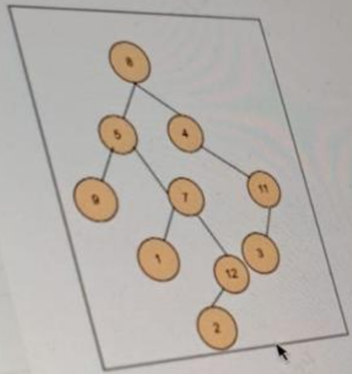
**Constraints:**
I) The length of the input string is less than 100.
II) Digits may be present in the input string.
III) The special characters that have to be removed from the given string are mentioned in the problem statement and everything else remains in the output string.

**Input Format:**
The first line of input is the user's spoken sentence.

**Output Format:**
The output consists of an intermediate string, part of the syntactical assessment.

QUESTION: **11** GROUP: **Technical** SECTION: **Data Structures and Algorithms**

Consider the given "Tree" and identify the degree of node 7




- 2
- 3
- None of the options given
- 4

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27 ☆

2
2
4

**Explanation 1:**
*The total number of printers available are 5.*
*For the first printer, T=4. The number of factors of integer '4' are 3 (1,2 and 4), which is printed in the first line of the output.*
*For the second printer, T=5. The number of factors of integer '5' are 2 (1 and 5), which is printed in the second line of the output.*
*Similarly, the number of pages that can be printed is found out for the following printers.*

**Sample Input 1:**

```
3
12
22
9
```

**Sample Output 1:**

```
6
4
3
```

**Explanation 2:**
*Total number of printers available are 3.*
*For the first printer, T=12. The number of factors of integer '12' are 6 (1,2,3,4,6 and 12), which is printed in the first line of the output.*
*For the second printer, T=22. the number of factors of integer '22' are 4 (1,2,11 and 22), which is printed in the second line of the output.*
*For the second printer, T=9. the number of factors of integer '9' are 3 (1,3 and 9), which is printed in the third line of the output.*

Current Code | Previous Versions | Select Language

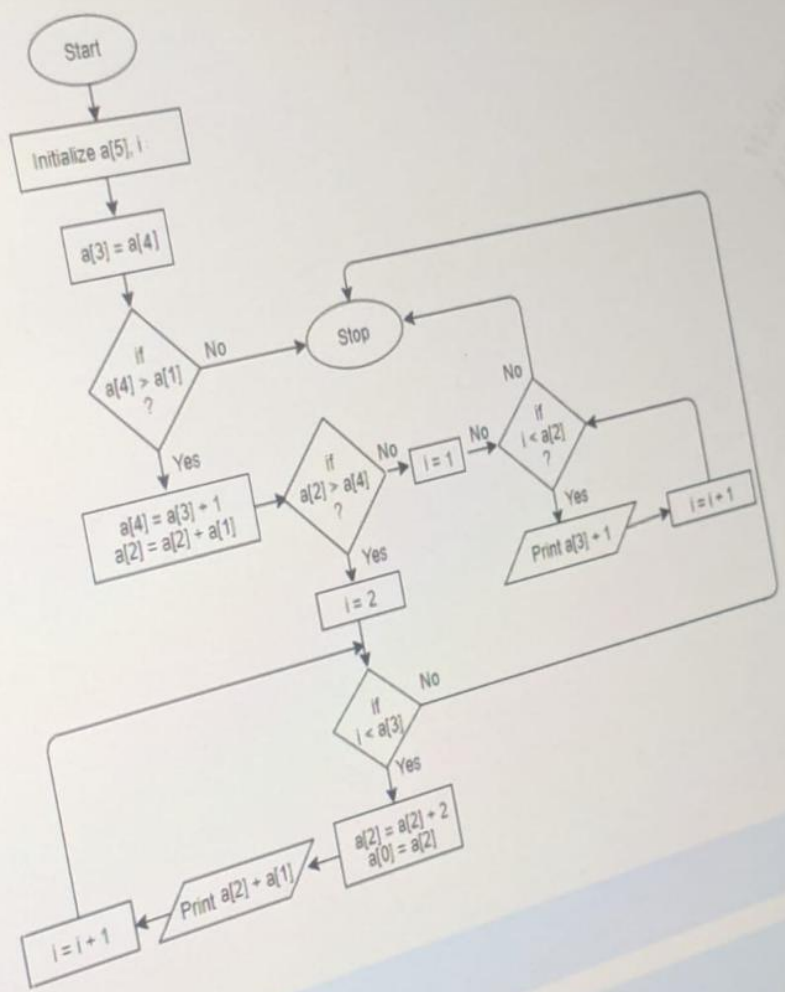
1

Previous Question | Next Question

ro

Instructions

Which set of array elements given in options will give the output as "15 17 19 21 23 25" if passed to the flowchart given below?

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5 2 9 1 2

3 4 4

Previous Question

Next

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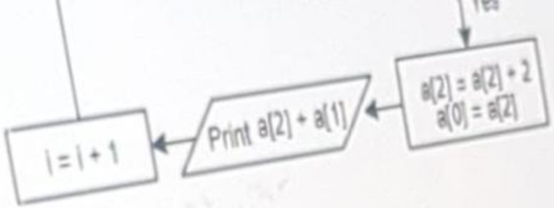


- ○ 5 2 9 1 2
- ○ 5 1 3 4 4
- ○ 6 1 8 2 1
- ○ 4 2 3 4 7

ePro

QUESTION 19 GROUP Technical SECTION OOPS & Solid Design Principles

Creational design patterns can be divided into:

I) class-creation patterns
II) object-creational patterns
III) methods-creational patterns

- Only (I) and (III)
- Only (II) and (III)
- Both (I) and (II)
- Only II

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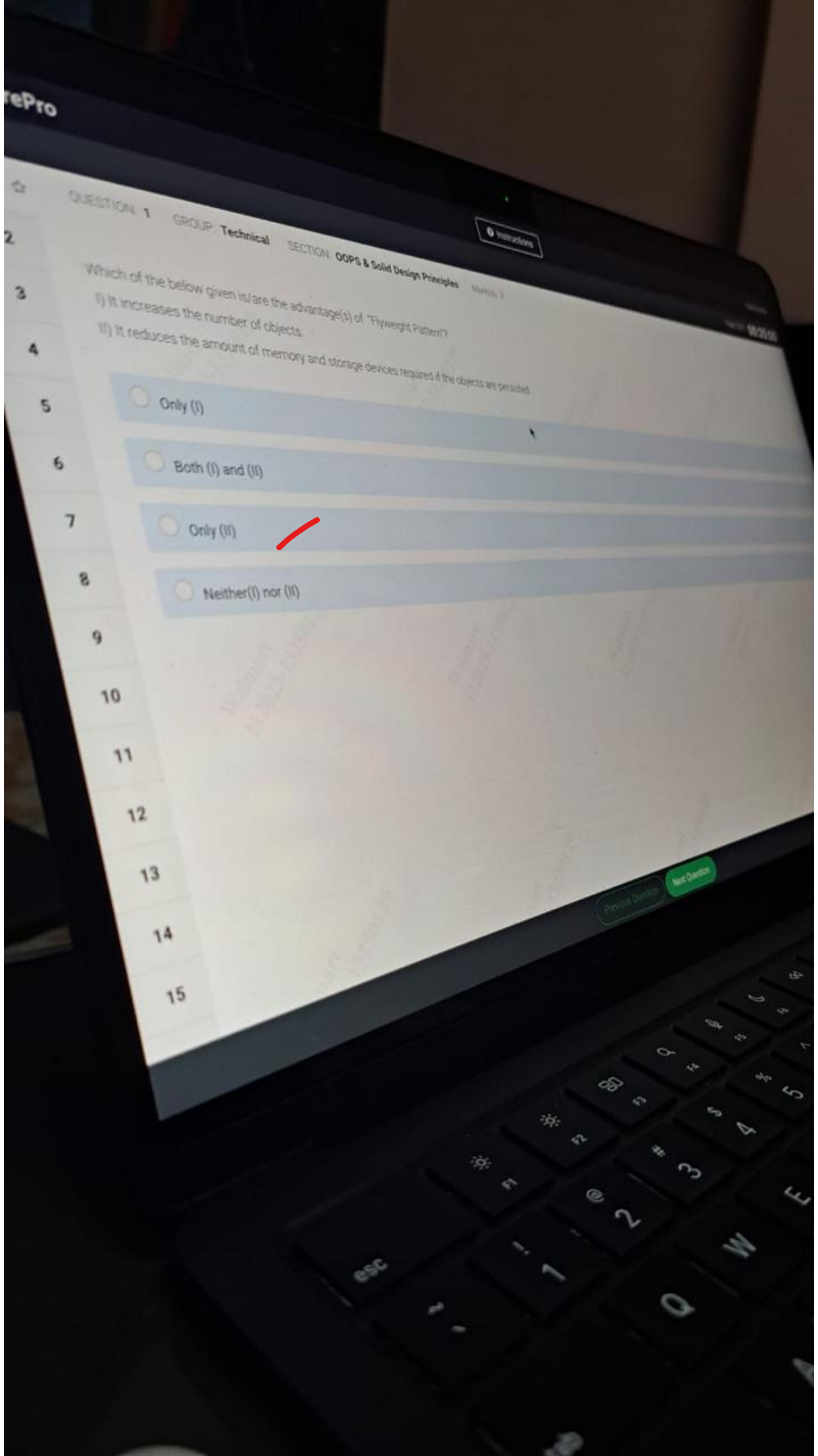
ePro
QUESTION: 1
GROUP: Technical
SECTION: OOPS & Solid Design Principles
Instructions
Which of the below given is/are the advantage(s) of "Flyweight Pattern"?
I) It increases the number of objects.
II) It reduces the amount of memory and storage devices required if the objects are persisted
Only (I)
Both (I) and (II)
Only (II)
Neither(I) nor (II)
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Previous Question
Next Question
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HirePro

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QUESTION: 9 GROUP: Technical SECTION: Data Structures and Algorithms

Consider the following function:

```
int Fun(struct Queue *Q)
{
    return((Q->rear + 1) % Q->capacity == Q->front);
}
```

What does the above function return:

- Size of the Queue
- "1" if the Queue is full
- Null
- "1" if the Queue is empty

ePro

QUESTION 8 GROUP Technical SECTION Operating Systems

Which one of the given options is correct for the below given operating systems?

I) Windows 10

II) Cent OS

III) Gentoo

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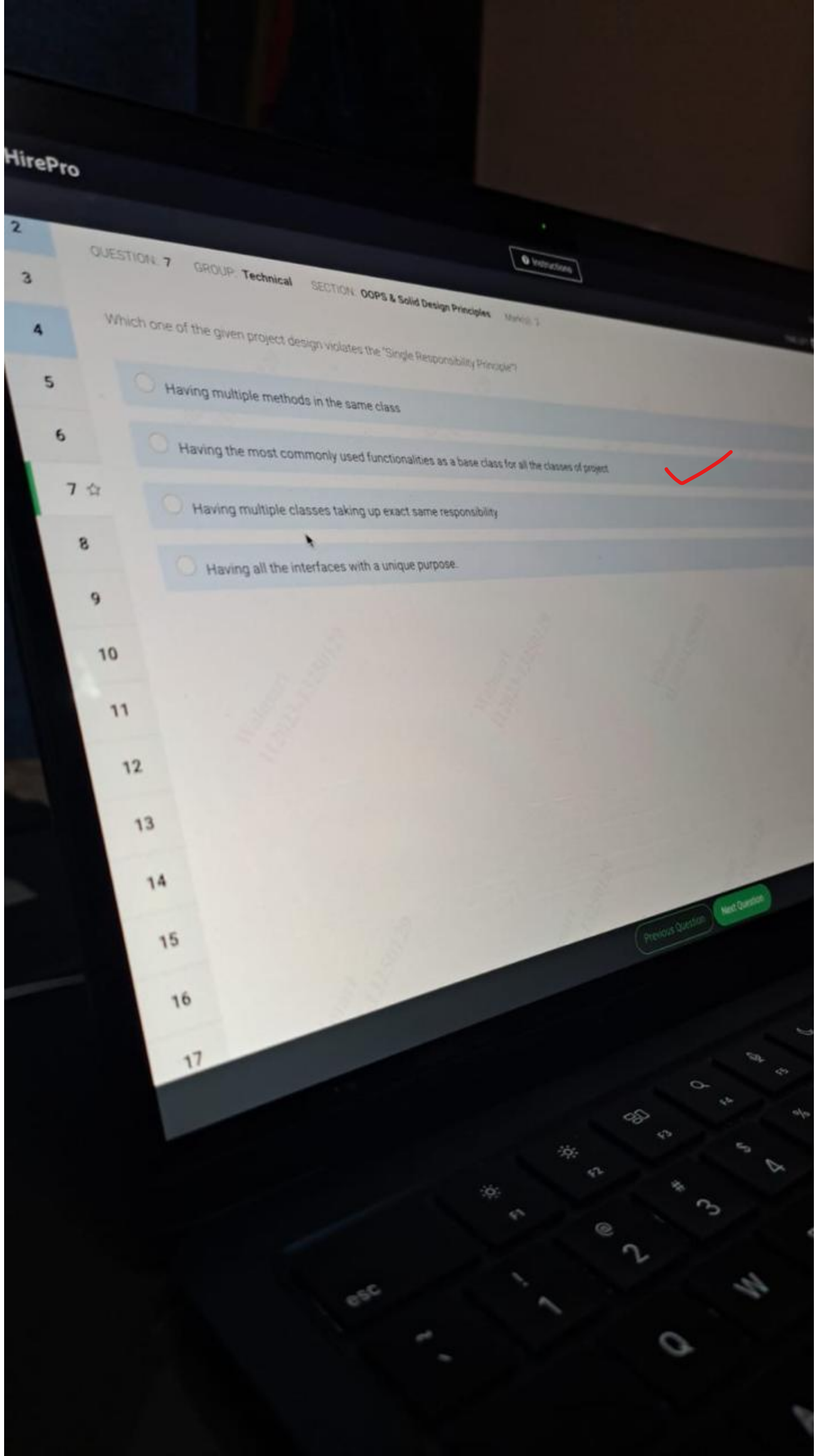
- Only (I) is a Linux distribution.
- Only (I) and (II) are Linux distributions.
- All (I), (II) and (III) are Linux distributions.
- Only (II) and (III) are Linux distributions.

Previous Question

Next Question


HirePro
Instructions
QUESTION: 7
GROUP: Technical
SECTION: OOPS & Solid Design Principles
Which one of the given project design violates the "Single Responsibility Principle"?
Having multiple methods in the same class
Having the most commonly used functionalities as a base class for all the classes of project
Having multiple classes taking up exact same responsibility
Having all the interfaces with a unique purpose.
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Previous Question
Next Question

Consider the table given below:

| STUDENT | | | | |
|---|---|---|---|---|
| STD_ID | STD_NAME | DATE_OF_BIRTH | PHONE | DEPT_NAME |
| 1 | aa | 10-10-1991 | 9988776655 | maths |
| 2 | bb | 11-09-1990 | 9977776655 | computers |
| 3 | cc | 02-09-1991 | 9977576655 | stats |
| 4 | dd | 02-09-1991 | 9977574655 | maths |

Which one of the query given below will display the names of students in the departments other than Maths and Compu

- SELECT std_name FROM student WHERE dept_name NOT EXISTS ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT EXISTS IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name IN ('maths','computers');

Instructions

QUESTION: 13 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

Which of the given below index is used to reorder the physical order of the table and search based on the key values?

- Unique Index
- Clustered Index
- NonClustered Index
- Function based index

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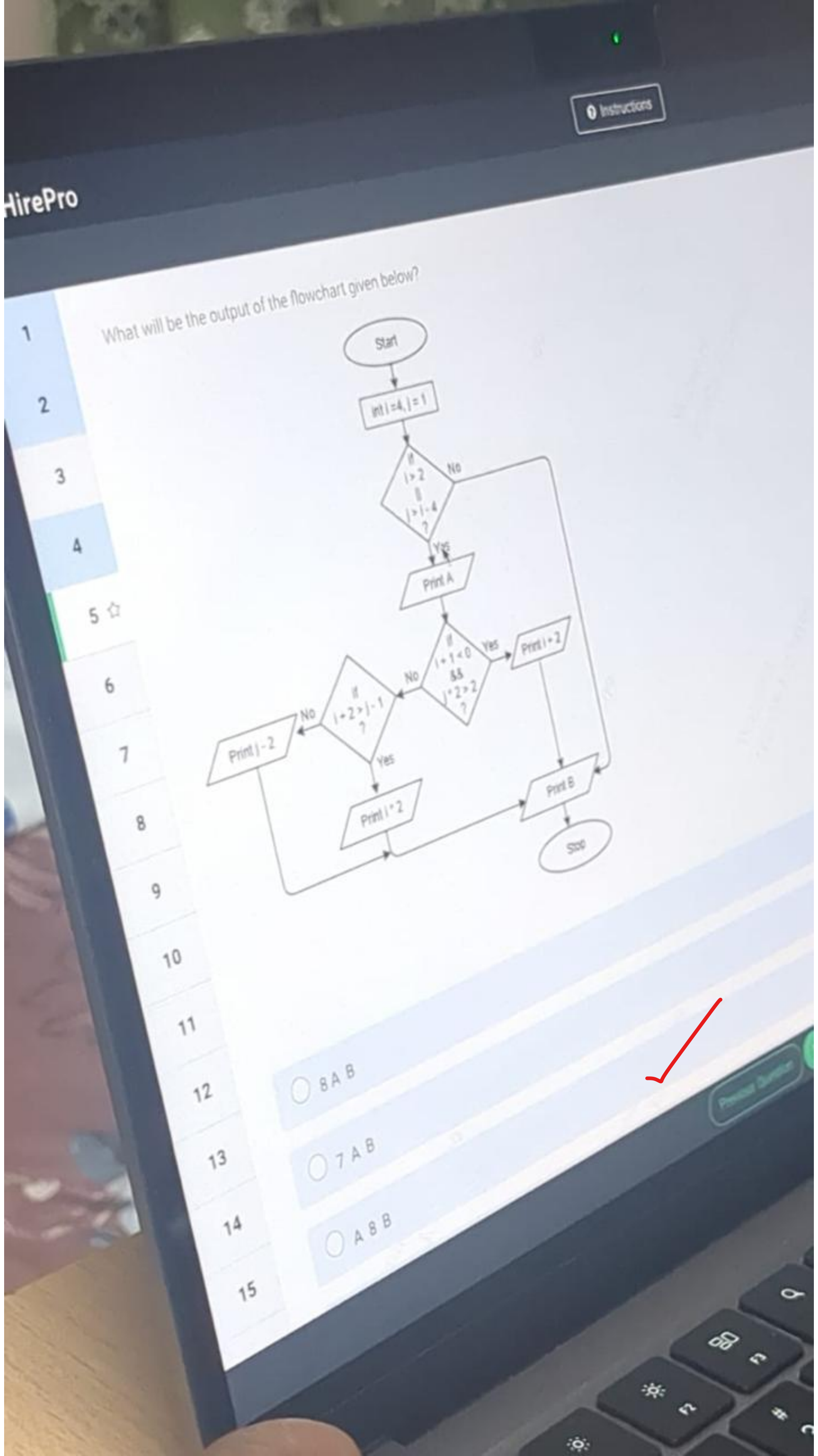
Previous Question

Next Question



Instructions
HirePro
What will be the output of the flowchart given below?
Start
int i =4, j = 1
if
i > 2
||
j > i - 4
?
No
Yes
Print A
if
i + 1 < 0
&&
j * 2 > 2
?
Yes
Print i + 2
No
if
i + 2 > j - 1
?
No
Print j - 2
Yes
Print i * 2
Print B
Stop
8 A B
7 A B
A 8 B
Previous Question
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Consider the following pseudo-code algorithm segment.

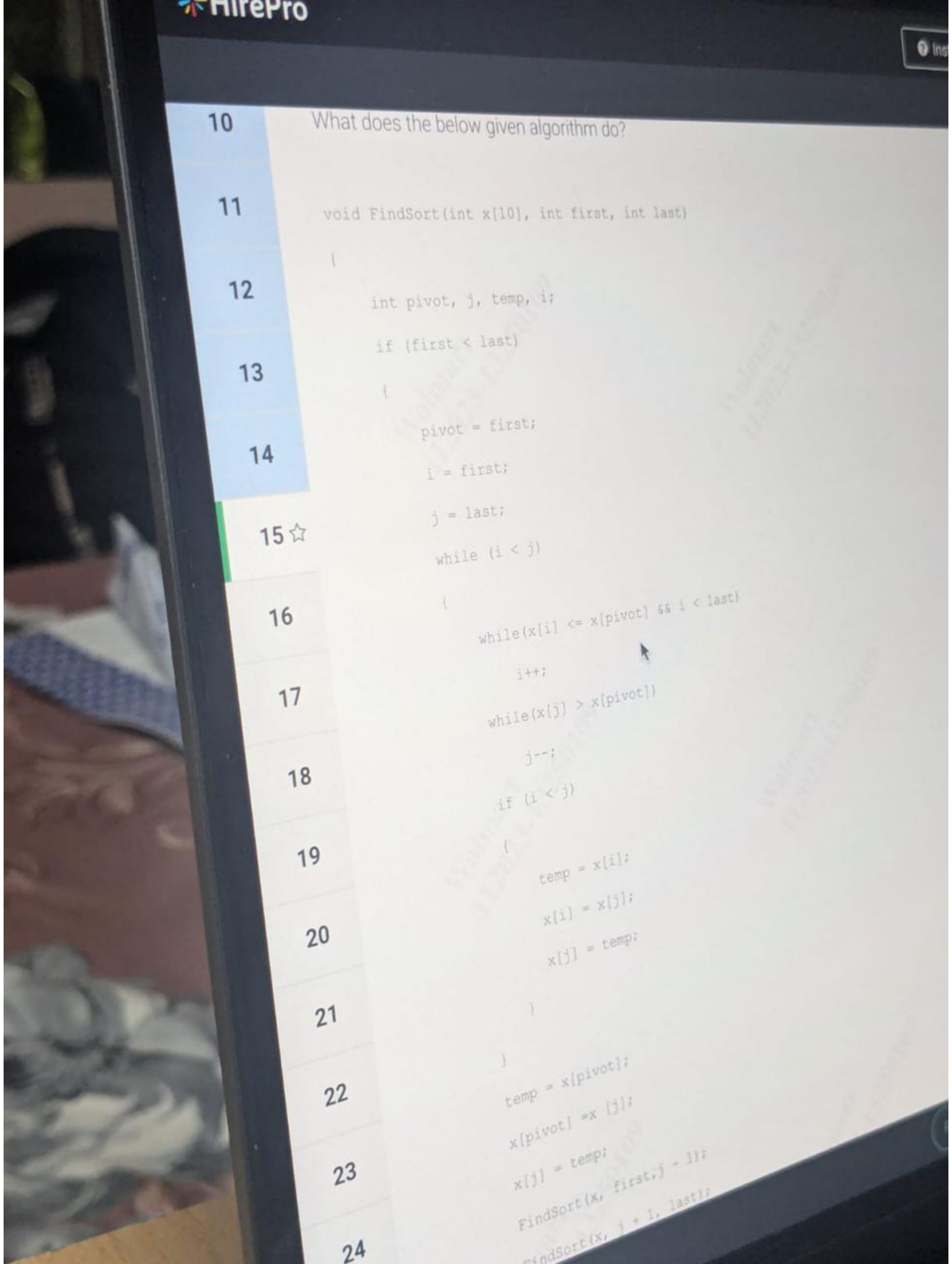
```
algo-1() {
    for all vertices u {
        num(u) = 0
        edges = null
    }
    i = 1
    while there is a vertex v such that num(v) == 0 {
        num(v) = i++
        enqueue(v)

        while queue is not empty {
            v = dequeue()
            for all vertices u adjacent to v {
                if num(u) == 0 {
                    num(u) = i++
                    enqueue(u)
                    attach edge (uv) to edges
                }
            }
        }
    }
}
```

What does the above algorithm describe?

- Depth first traversal
- Breadth first traversal
- Array based implementation of queue
- Deleting nodes from a queue

16 17 18 19 20 21 22 23 24 25

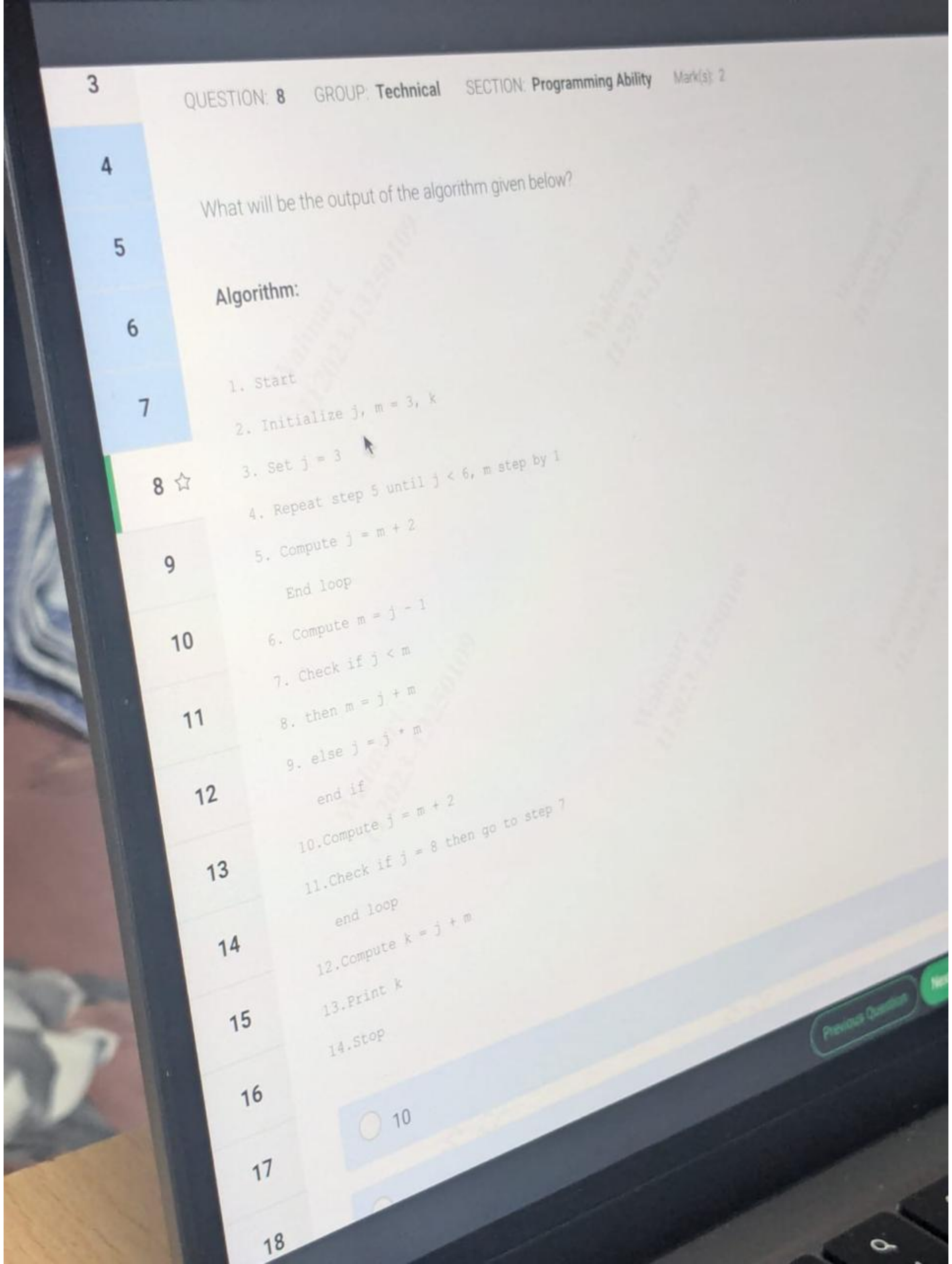
Previous Question Next Question




What does the below given algorithm do?

```
void FindSort(int x[10], int first, int last)
{
    int pivot, j, temp, i;
    if (first < last)
    {
        pivot = first;
        i = first;
        j = last;
        while (i < j)
        {
            while(x[i] <= x[pivot] && i < last)
                i++;
            while(x[j] > x[pivot])
                j--;
            if (i < j)
            {
                temp = x[i];
                x[i] = x[j];
                x[j] = temp;
            }
        }
        temp = x[pivot];
        x[pivot] =x [j];
        x[j] = temp;
        FindSort(x, first,j - 1);
        FindSort(x, j + 1, last);
```

Quick Sort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QUESTION: 8 GROUP: Technical SECTION: Programming Ability Mark(s): 2
What will be the output of the algorithm given below?
Algorithm:
1. Start
2. Initialize j, m = 3, k
3. Set j = 3
4. Repeat step 5 until j < 6, m step by 1
5. Compute j = m + 2
End loop
6. Compute m = j - 1
7. Check if j < m
8. then m = j + m
9. else j = j * m
end if
10.Compute j = m + 2
11.Check if j = 8 then go to step 7
end loop
12.Compute k = j + m
13.Print k
14.Stop
10
Previous Question

Ans : 12

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```
TOWER(N, BEG, AUX, END)
 If N = 0 Then
    Return
  End - If
  ____________
      Write BEG  END
      Call TOWER(N - 1, BEG, END, AUX) and TEMP  = BEG, BEG  = AUX, AUX  = TEMP
   End - For
   Return
```

- For N to 2
- For N - 1 to 1
- For N to 1
- For N to N - 1

Previous Question

For the first printer, T=4. The number of factors of integer ...
For the second printer, T=5. The number of factors of integer '5' are 2 (1 and 5), which is printed in the second ...
Similarly, the number of pages that can be printed is found out for the following printers.

**Sample Input 1:**

```
3
12
22
9
```

**Sample Output 1:**

```
6
4
3
```

**Explanation 2:**

Total number of printers available are 3.
For the first printer, T=12. The number of factors of integer '12' are 6 (1,2,3,4,6 and 12), which is printed in the first line of the output.
For the second printer, T=22. the number of factors of integer '22' are 4 (1,2,11 and 22), which is printed in the second line of the output.
For the second printer, T=9. the number of factors of integer '9' are 3 (1,3 and 9), which is printed in the third line of the output.

Select Language

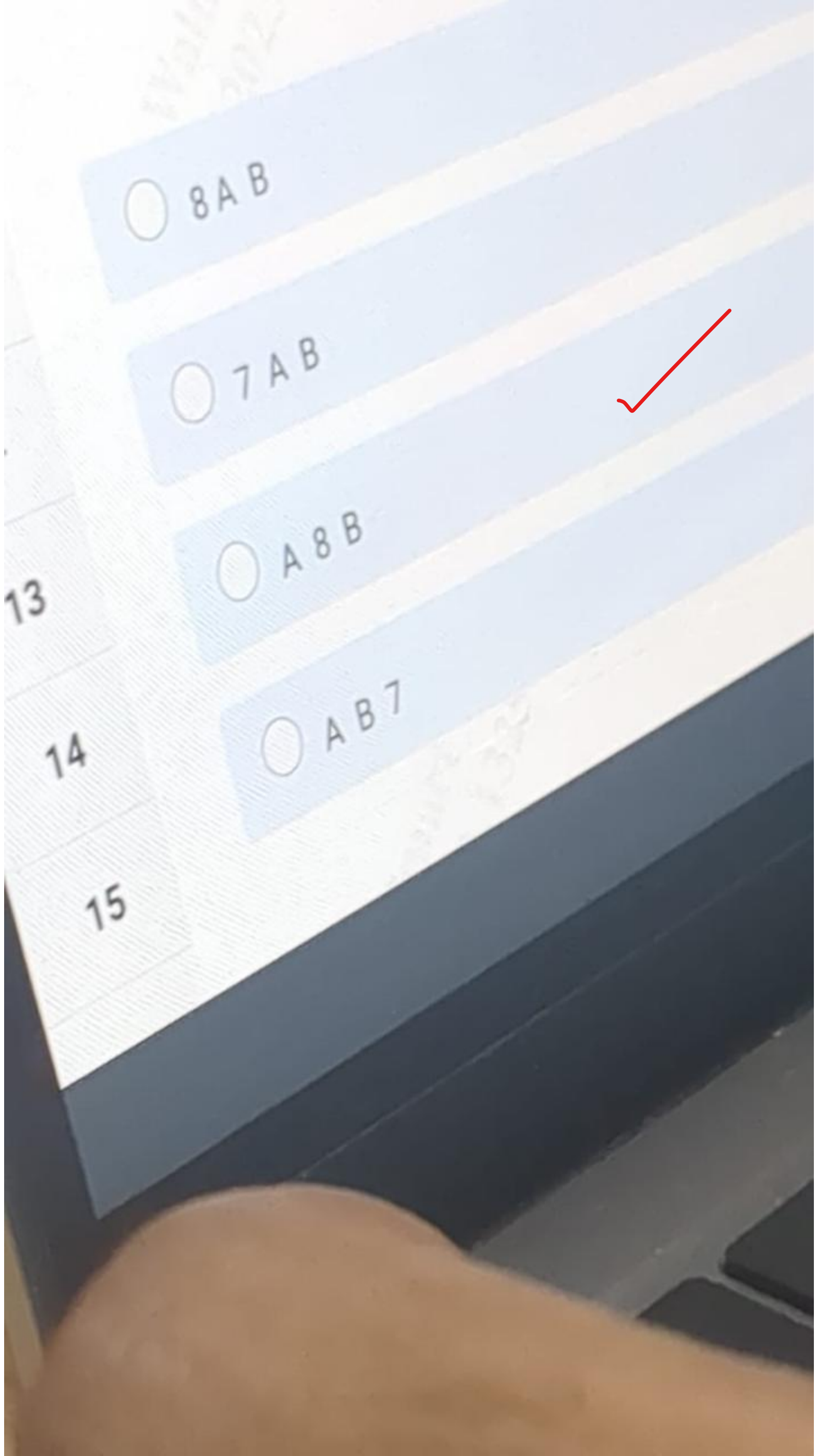
Current Code Previous Versions

```
1
```

Previous Question Next Question

```
    {
        temp = x[i];
        x[i] = x[j];
        x[j] = temp;
    }
  }
  temp = x[pivot];
  x[pivot] =x [j];
  x[j] = temp;
  FindSort(x, first,j - 1);
  FindSort(x, j + 1, last);
 }
}
```

- ○ Quick sort implementation using an array in ascending order.
- ○ Bucket sort implementation using an array in ascending order.
- ○ Heap sort implementation using an array in ascending order.
- ○ Selection sort implementation using an array in ascending order.


8 A B
7 A B
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A 8 B
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A B 7
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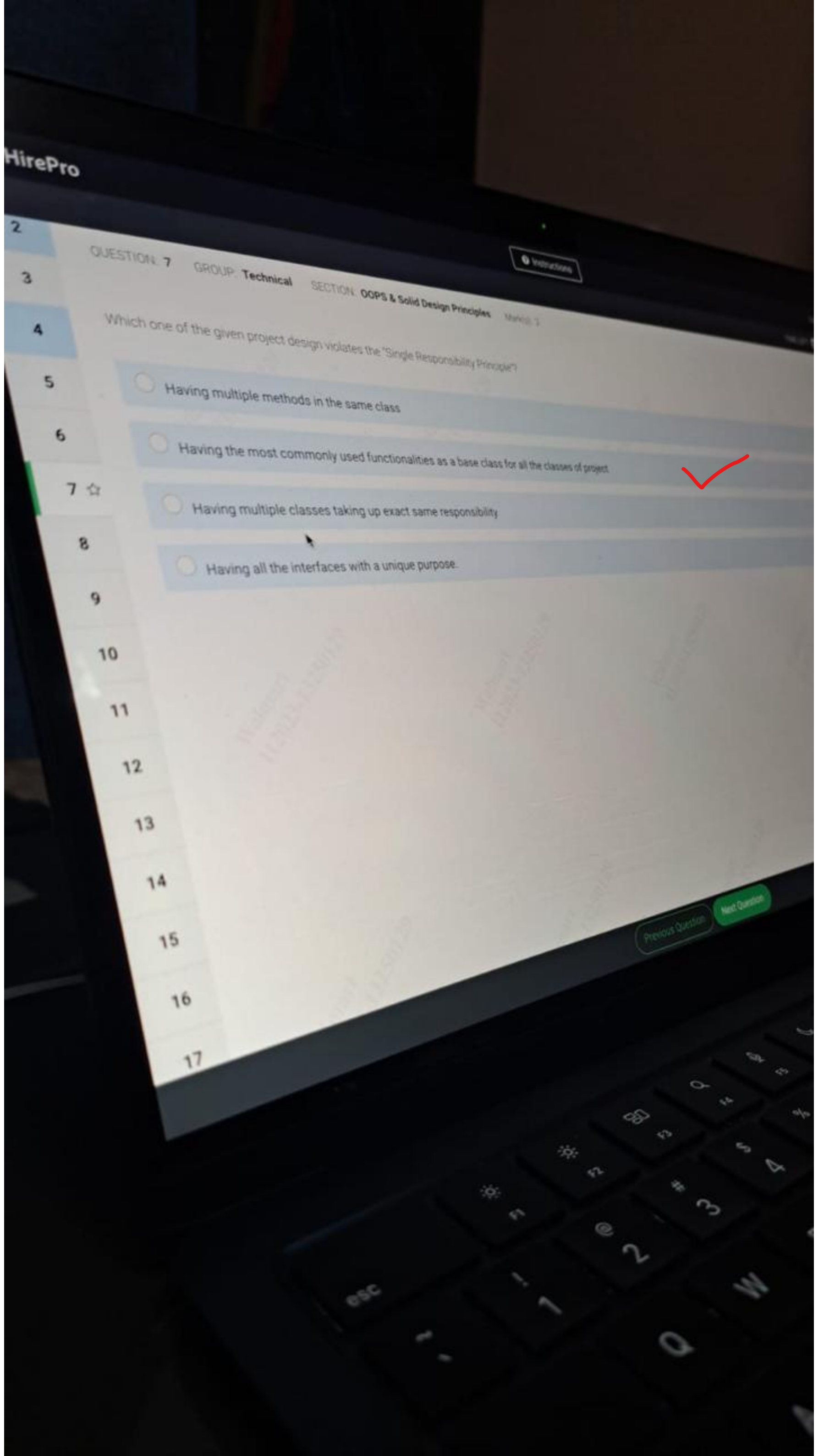
HirePro
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QUESTION: 7
GROUP: Technical
SECTION: OOPS & Solid Design Principles
Instructions
Which one of the given project design violates the "Single Responsibility Principle"?
Having multiple methods in the same class
Having the most commonly used functionalities as a base class for all the classes of project
Having multiple classes taking up exact same responsibility
Having all the interfaces with a unique purpose.
Previous Question
Next Question

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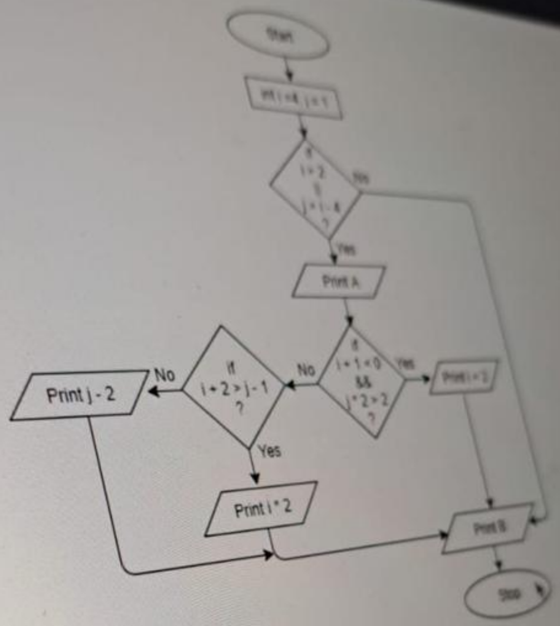
What will be the output of the flowchart given below?

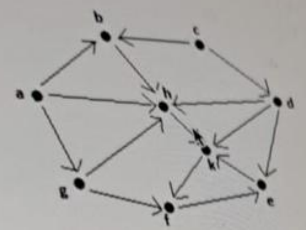



- 8 A B
- A B 7
- 7 A B
- A 8 B

ePro

QUESTION: 10 GROUP: Technical SECTION: Data Structures and Algorithms

Consider the given graph:




Which one of the vertex given in options has the maximum in-degree?

- h
- d
- f
- a

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QUESTION: 21 GROUP: Technical SECTION: Networking Mark(s): 2

Which layer divides each message into packets at the source and re-assembles them at the destination?

- Transport layer
- Physical layer
- Data link layer
- Network layer

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QUESTION 18 GROUP Technical SECTION OOPS & Solid Design Principles Marks: 1

Instructions

Suppose a class has public visibility. In this class we define a protected method. Which one of the statement given in options is correct?

- From within protected methods you do not have access to public methods.
- This method is accessible from within the class itself and from within all classes defined in the same package as the class itself.
- This method is only accessible from inside the class itself and from inside all subclasses.
- In a class, you cannot declare methods with a lower visibility than the visibility of the class in which it is defined.

ePro

QUESTION: 17 GROUP: Technical SECTION: Data Structures and Algorithms

What will be the output of the given algorithm for a = 2 and b = 5?
[**Note:** ^ is XOR operator]

**Algorithm:**

```
1) Start
2) Initialize a and b
3) Read a and b
4) Calculate a ^= b
             b ^= a
             a ^= b
5) Print a and b
6) Stop
```

- 5 and 5
- 0 and 1
- 5 and 2
- 0 and 0

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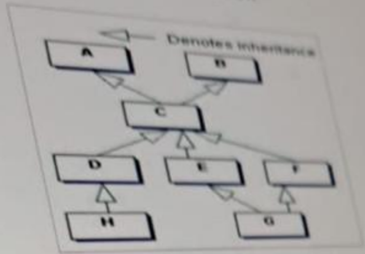
QUESTION: 14 GROUP: Technical SECTION: OOPS & Solid Design Principles

Consider the figure given below




Referring to the above diagram, which one of the statement given in options is true?

- Class A and B have same non-private data members.
- Class A and C have same non-private data members.
- Class C has all non private data members of class A.
- Class B and C have same non-private data members.

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14 ☆
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QUESTION: 23 GROUP: Technical SECTION: RDBMS and SQL

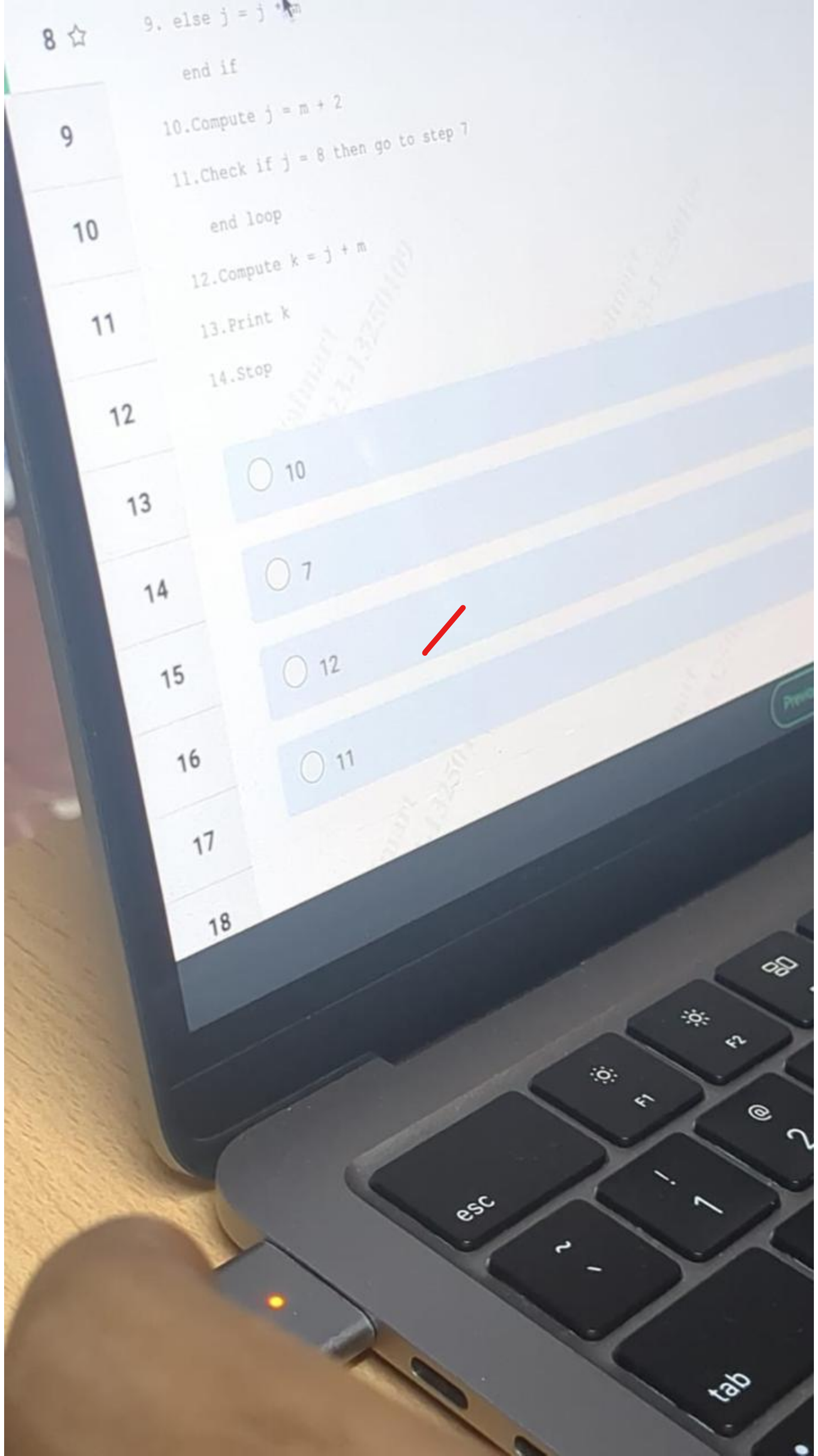
Consider the table given below

| Product_Info | | |
|---|---|---|
| **P_Id** | **P_Name** | **Cost** |
| 1001 | Container | 200 |
| 1002 | Lunch box | 450 |
| 1003 | Signoware | 650 |
| 1005 | Tupperware | 800 |

What will be the output of the query given below?

```
SELECT AVG(Cost) from Product_Info;
```

- 530
- 600
- 525
- 425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9. else j = j * m
end if
10.Compute j = m + 2
11.Check if j = 8 then go to step 7
end loop
12.Compute k = j + m
13.Print k
14.Stop
10
7
12
11
esc
F1
F2
tab

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Consider the following pseudo-code algorithm segment.

```
algo-1(){
    for all vertices u {
        num(u) = 0
        edges = null
    }
    i = 1
    while there is a vertex v such that num(v) == 0 {
        num(v) = i++
        enqueue(v)

        while queue is not empty {
            v = dequeue()
            for all vertices u adjacent to v {
                if num(u) == 0 {
                    num(u) = i++
                    enqueue(u)
                    attach edge (uv) to edges
                }
            }
        }
    }
}
```

What does the above algorithm describe?

- Deleting nodes from a queue
- Breadth first traversal
- Array based implementation of queue
- Depth first traversal

Previous Question

ePro

Instructions

Abhishek Maurya

TIME LEFT: 00:39:38

QUESTION: 26 GROUP: Coding SECTION: Coding 1 Mark(s): 20

A group of users on a network want access to a printer for printing a certain number of pages T. This network is connected to a group of printers. These printers have limited ink and due to this, the printers have been set to print only t pages. Here, t pages will be calculated as the total number of factors of the number of pages T specified by the user.

The number of printers that are available (N) and the number of pages (T) is specified by the user.
Zero-indexed array means an array whose index starts from 0. Hence, the first element's index is 0; the second element's index is 1, the kth element's index is (k-1) and so on.

Write a program to determine the number of pages that can be printed from each printer A[i] such that 0 ≤ i < N.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

**Constraints:**
I) 1 <= N <= 50000
II) 1 < A[i] < 100000

**Input Format:**
The first line contains the number of printers N.
The next N lines contain the total number of pages T to be printed that are given by the user.

**Output Format:**
N lines of output contain t, number of pages selected for printing for each printer.

**Sample Input 1:**

Instead of using two recursive calls for the solution to the Tower Of Hanoi problem, only one

**Input :**

N, BEG, AUX, END: Number of pegs, beginning pole, auxiliary pole, ending pole

**Output :**

Arranging pegs in ascending order from the Beginning pole to End pole

```
TOWER(N, BEG, AUX, END)
 If N = 0 Then
    Return
  End - If
  ____________
      Write BEG  END
      Call TOWER(N - 1, BEG, END, AUX) and TEMP  = BEG, BEG  = AUX, AUX  = TEMP
    End - For
    Return
```

- For N to 2
- For N to N - 1
- For N - 1 to 1
- For N to 1

16
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27 ☆

**Constraints:**

I) The length of the input string is less than 100.
II) Digits may be present in the input string.
III) The special characters that have to be removed from the given string are mentioned in the problem statement and everything else remains in the output string

**Input Format:**

The first line of input is the user's spoken sentence.

**Output Format:**

The output consists of an intermediate string, part of the syntactical assessment.

**Sample Input1:**

Barbara likes this

**Sample Output1:**

Brbaraalkisethsi

**Explanation1:**

The input string is " Barbara likes this".
Exchanging vowels and consonants, if a vowel is followed by a consonant gives " Brbaraa lkise thsi ".
Removing spaces give " Brbaraalkisethsi ", which is printed as the output.

**Sample Input2:**

You are 2 g00d @ coding.

**Sample Output2:**

Yourae2g00dcdongi.

**Explanation 2:**

The input string is " You are 2 g00d @ coding. "
Exchanging vowels and consonants, if a vowel is followed by a consonant gives " You rae 2 g00d @ cdongi. "
Removing spaces gives " Yourae2g00d@cdongi."
Removing special characters gives " Yourae2g00dcdongi.", which is printed as output.

Select Language

End Test

Previous Versions

27

A group of users on a network want access to a printer for printing a certain number of pages T. This network is connected to a group of printers. These printers have limited ink and due to this, the printers have been set to print only t pages. Here, t pages will be calculated as the total number of factors of the number of pages T specified by the user.

The number of printers that are available (N) and the number of pages (T) is specified by the user.
Zero-indexed array means an array whose index starts from 0. Hence, the first element's index is 0, the second element's index is 1, the kth element's index is (k-1) and so on.

Write a program to determine the number of pages that can be printed from each printer A[i] such that 0 ≤ i < N.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

**Constraints:**
I) 1 <= N <= 50000
II) 1 < A[i] < 100000

**Input Format:**
The first line contains the number of printers N.
The next N lines contain the total number of pages T to be printed that are given by the user.

**Output Format:**
N lines of output contain t, number of pages selected for printing for each printer.

**Sample Input 1:**

```
5
4
5
2
3
6
```

**Sample Output 1:**

```
3
2
2
2
4
```

**Explanation 1:**

Previous Question Next Question

Instructions

QUESTION: 24 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

Consider the tables given below:

**STUDENTS**

| ROLLNO | NAME | PIN |
|---|---|---|
| 5780 | kritika | 761005 |
| 5733 | Prakash | 560037 |
| 5767 | Ranjan | 500038 |
| 7739 | Nabeen | 560037 |
| 5789 | Sanjay | 560037 |
| 5797 | Janifer | 751003 |
| 5763 | Resma | 500038 |
| 5776 | Jeebitesh | 751003 |

**CITY**

| PIN | NAME | POPULATION |
|---|---|---|
| 761005 | Brahmapur | 355827 |
| 560037 | Bengalore | 11556907 |
| 500038 | Hyderabad | 11723548 |
| 751003 | Bhubaneswar | 886397 |

Which of the query/queries given below will retrieve the students whose city's population is more than 11lakhs?

```
I) SELECT s.name FROM students s, city c WHERE c.pin=s.pin and c.population>1100000;
II) SELECT name FROM students WHERE pin IN (SELECT pin FROM city WHERE population>1100000);
III) SELECT s.name FROM students s right outer join city c WHERE c.pin=s.pin and c.population>1100000;
```

- Only (I) and (II)
- Only (II)
- All (I), (II) and (III)
- Only (I)

Previous Question Next Question

Instructions

QUESTION 4 GROUP Technical SECTION **OOPS & Solid Design Principles** Mark(s): 2

Suppose a class has public visibility. In this class we define a protected method. Which one of the statement given in options is correct?

- In a class, you cannot declare methods with a lower visibility than the visibility of the class in which it is defined.
- This method is accessible from within the class itself and from within all classes defined in the same package as the class itself.
- From within protected methods you do not have access to public methods.
- This method is only accessible from inside the class itself and from inside all subclasses.

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Previous Question

Next Question

QUESTION: 1 GROUP: Technical SECTION: Operating Systems

Which one of the given options is correct for the below given procedures?

NO GLOBAL VAR

```
int temp;
int ab()
{
        return 7;
    }

    void abc()
    {
        temp = ab();
        printf("%d", temp);
    }
```

- Only procedure ab() is re-entrant.
- Neither procedure ab() nor abc() is re-entrant.
- Procedures ab() and abc() are re-entrant.
- Only procedure abc() is re-entrant.

6 7 8 9 10 11 12 13 14

Instructions

QUESTION: 3 GROUP: **Technical** SECTION: **Networking** Mark(s): 2

Programmer is working on a program to convert data into an unreadable format. The program should also convert back the data into

programmer use?

- Risk Analysis
- Digital Signature
- Hashing
- Encryption and Decryption ✓

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HirePro

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Consider the following pseudo-code algorithm segment.

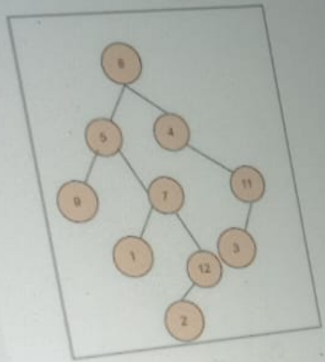
```
algo-1() {
    for all vertices u {
        num(u) = 0
        edges = null
    }
    i = 1
    while there is a vertex v such that num(v) == 0 {
        num(v) = i++
        enqueue(v)

        while queue is not empty {
            v = dequeue()
            for all vertices u adjacent to v {
                if num(u) == 0 {
                    num(u) = i++
                    enqueue(u)
                    attach edge (uv) to edges
                }
            }
        }
    }
}
```

What does the above algorithm describe?

- Depth first traversal
- Deleting nodes from a queue
- Breadth first traversal

Consider the given "Tree" and identify the degree of node 7




- 4
- 2
- 3
- None of the options given

QUESTION 5 GROUP Technical SECTION Data Structures and Algorithms

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16

Instead of using two recursive calls for the solution to the Tower Of Hanoi problem, only one recursive call can be used. What would come

**Input :**

N, BEG, AUX, END: Number of pegs, beginning pole, auxiliary pole, ending pole

**Output :**

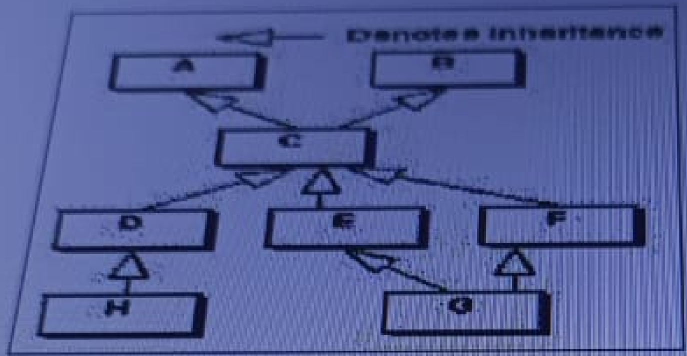
Arranging pegs in ascending order from the Beginning pole to End pole

```
TOWER(N, BEG, AUX, END)
 If N = 0 Then
    Return
 End - If
 ____________
     Write BEG  END
     Call TOWER(N - 1, BEG, END, AUX) and TEMP = BEG, BEG = AUX, AUX = TEMP
   End - For
   Return
```

- For N to 2
- For N to N - 1
- For N - 1 to 1

2 ☆

Consider the figure given below:






Referring to the above diagram, which one of the statement given in options is true?

- Class A and C have same non-private data members.
- Class B and C have same non-private data members.
- Class C has all non private data members of class A.
- Class A and B have same non-private data members.

QUESTION 3 GROUP Technical SECTION OOPS & Solid Design Principles Mark(s): 2

1
2
3
4

Which of the below given is/are the advantage(s) of "Flyweight Pattern"?

I) It increases the number of objects.

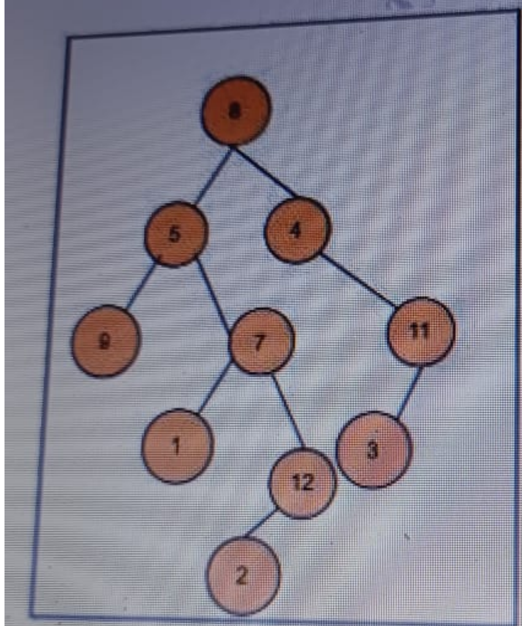
II) It reduces the amount of memory and storage devices required if the objects are persisted

- Both (I) and (II)
- Neither(I) nor (II)
- Only (II)
- Only (I)

ePro

QUESTION: 1 GROUP: Technical SECTION: Data Struc

Consider the given "Tree" and identify the degree of node 7




2

4

None of the options given

QUESTION 5 GROUP Technical SECTION OOPS & Solid Design Principles Mark(s): 1

Which of the below given statement(s) is/are true?

I) Variables declared inside a method have method level scope and can't be accessed outside the method.
II) A variable declared inside pair of brackets "{" and "}" in a method has scope withing the brackets only.
III) Any variable defined in a class outside of any method cannot be used by all member methods.
IV) For a variable to be read after the termination of a loop, it must be declared before the body of the loop.

- Only (I), (II) and (IV)
- Only (II)
- Only (III)
- Only (I), (II) and (III)

QUESTION 4 GROUP Technical SECTION RDBMS and SQL Mark(s): 2

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10

Consider the table given below:

| Product_Info | | |
|---|---|---|
| P_Id | P_Name | Cost |
| 1001 | Container | 200 |
| 1002 | Lunch box | 450 |
| 1003 | Signoware | 650 |
| 1005 | Tupperware | 800 |



What will be the output of the query given below?

```
SELECT AVG(Cost) from Product_Info;
```

- 525
- 600

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7 ☆
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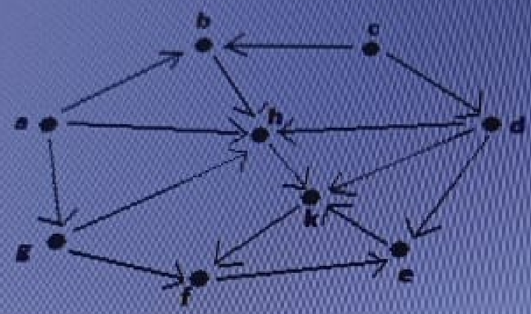
QUESTION: 7 GROUP: **Technical** SECTION: **Data Structures and Algorithms** Mark(s): 2

Consider the given graph:






Which one of the vertex given in options has the maximum in-degree?

- h
- d
- a
- f

3 4 5 6 7 8 9 10 11 12 13 14

**Algorithm:**

```
1. Start
2. Initialize j, m = 3, k
3. Set j = 3
4. Repeat step 5 until j < 6, m step by 1
5. Compute j = m + 2
   End loop
6. Compute m = j - 1
7. Check if j < m
8. then m = j + m
9. else j = j * m
   end if
10.Compute j = m + 2
11.Check if j = 8 then go to step 7
   end loop
12.Compute k = j + m
13.Print k
14.Stop
```

1 2 3 4 5 6 ☆ 7 8 9 10 11 12

Consider the following function:

```
int Fun(struct Queue* Q)
{
    return ((Q->rear + 1) % Q->capacity == Q->front);
}
```



What does the above function return:

- "1" if the Queue is empty
- Size of the Queue
- "1" if the Queue is full
- Null

QUESTION: **11** GROUP: **Technical** SECTION: **Networking** Mark(s): 2

Which layer divides each message into packets at the source and re-assembles them at the destination?

- Physical layer
- Network layer
- Transport layer
- Data link layer

QUESTION: 9 GROUP: **Technical** SECTION: **Programming Ability** Mark(s): 2

Consider the following pseudo-code algorithm segment.

```
algo-1(){
    for all vertices u {
        num(u) = 0
        edges = null
    }
    i = 1
    while there is a vertex v such that num(v) == 0 {
        num(v) = i++
        enqueue(v)

        while queue is not empty {
            v = dequeue()
            for all vertices u adjacent to v {
                if num(u) == 0 {
                    num(u) = i++
                    enqueue(u)
                    attach edge (uv) to edges
                }
            }
        }
    }
}
```

What does the above algorithm describe?

HirePro

QUESTION 13 GROUP Technical SECTION Data Structures and Algorithms Mark(s): 2

What will be the output of the given algorithm for a = 2 and b = 5?

[Note: ^ is XOR operator]



**Algorithm:**

```
1) Start
2) Initialize a and b
3) Read a and b
4) Calculate a ^= b
             b ^= a
             a ^= b
5) Print a and b
6) Stop
```



5 2

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0 and 1

HirePro

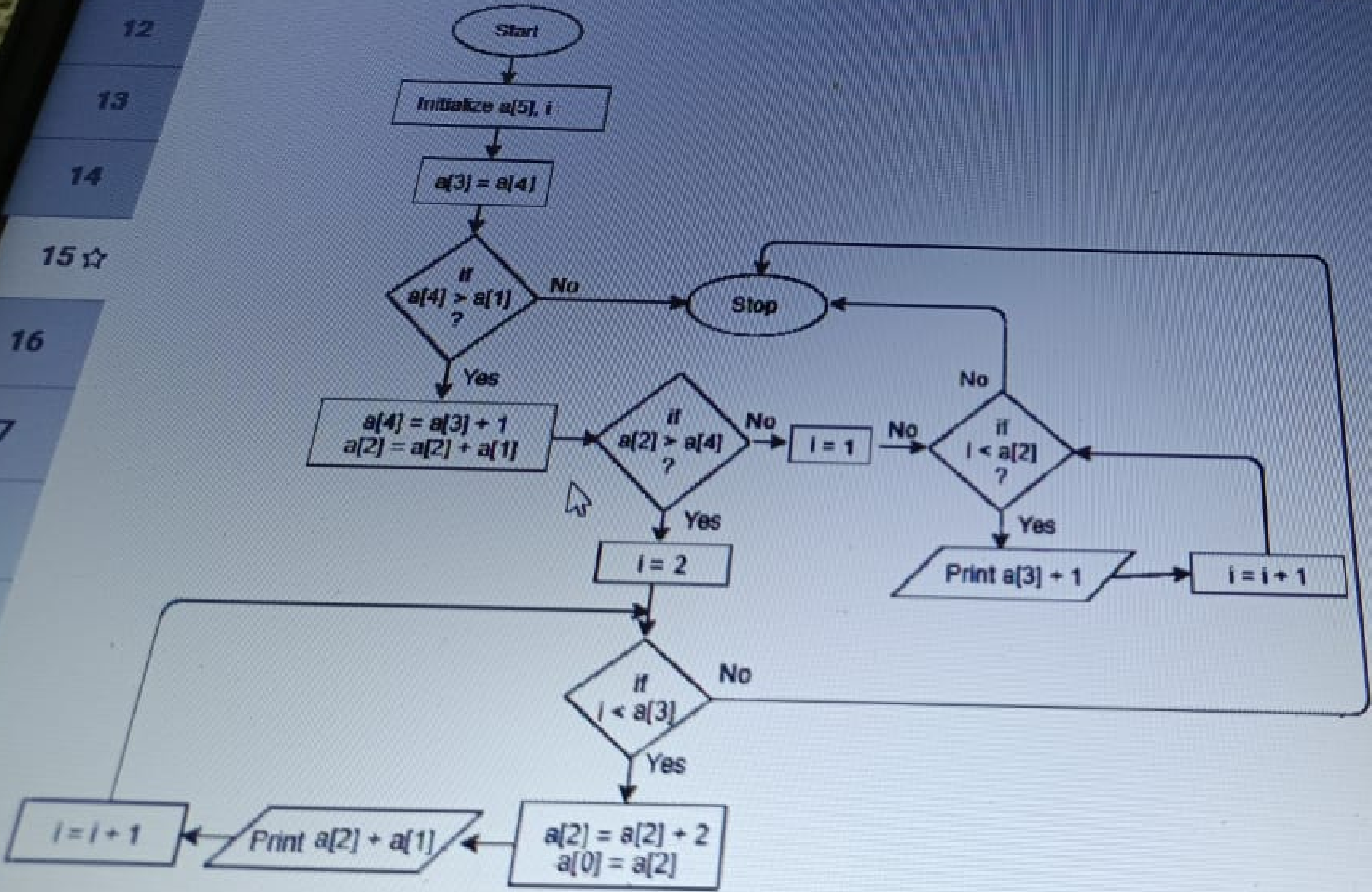
QUESTION: 12 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

Suppose a class has public visibility. In this class we define a protected method. Which one of the statement given in options is correct?

- From within protected methods you do not have access to public methods.
- This method is accessible from within the class itself and from within all classes defined in the same package as the class itself.
- In a class, you cannot declare methods with a lower visibility than the visibility of the class in which it is defined.
- This method is only accessible from inside the class itself and from inside all subclasses.

QUESTION: 15 GROUP: Technical SECTION: Programming Ability Mark(s): 2

Which set of array elements given in options will give the output as "15 17 19 21 23 25" if passed to the flowchart given below?






?

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QUESTION 16 GROUP Technical SECTION OOPS & Solid Design Principles Mark(s): 7

Creational design patterns can be divided into

I) class-creation patterns
II) object-creational patterns
III) methods-creational patterns

- Both (I) and (II)
- Only II
- Only (II) and (III)
- Only (I) and (III)

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14 ☆
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Instead of using two recursive calls for the solution to the Tower Of Hanoi problem, only one recursive call can be used. What would come in the blank spaces to complete the algorithm using only one recursive call?

**Input :**

N, BEG, AUX, END: Number of pegs, beginning pole, auxiliary pole, ending pole

**Output :**

Arranging pegs in ascending order from the Beginning pole to End pole.

```
TOWER(N, BEG, AUX, END)
  If N = 0 Then
    Return
  End - If
  ____________

  Write BEG  END
  Call TOWER(N - 1, BEG, END, AUX) and TEMP = BEG, BEG = AUX, AUX = TEMP
End - For
Return
```

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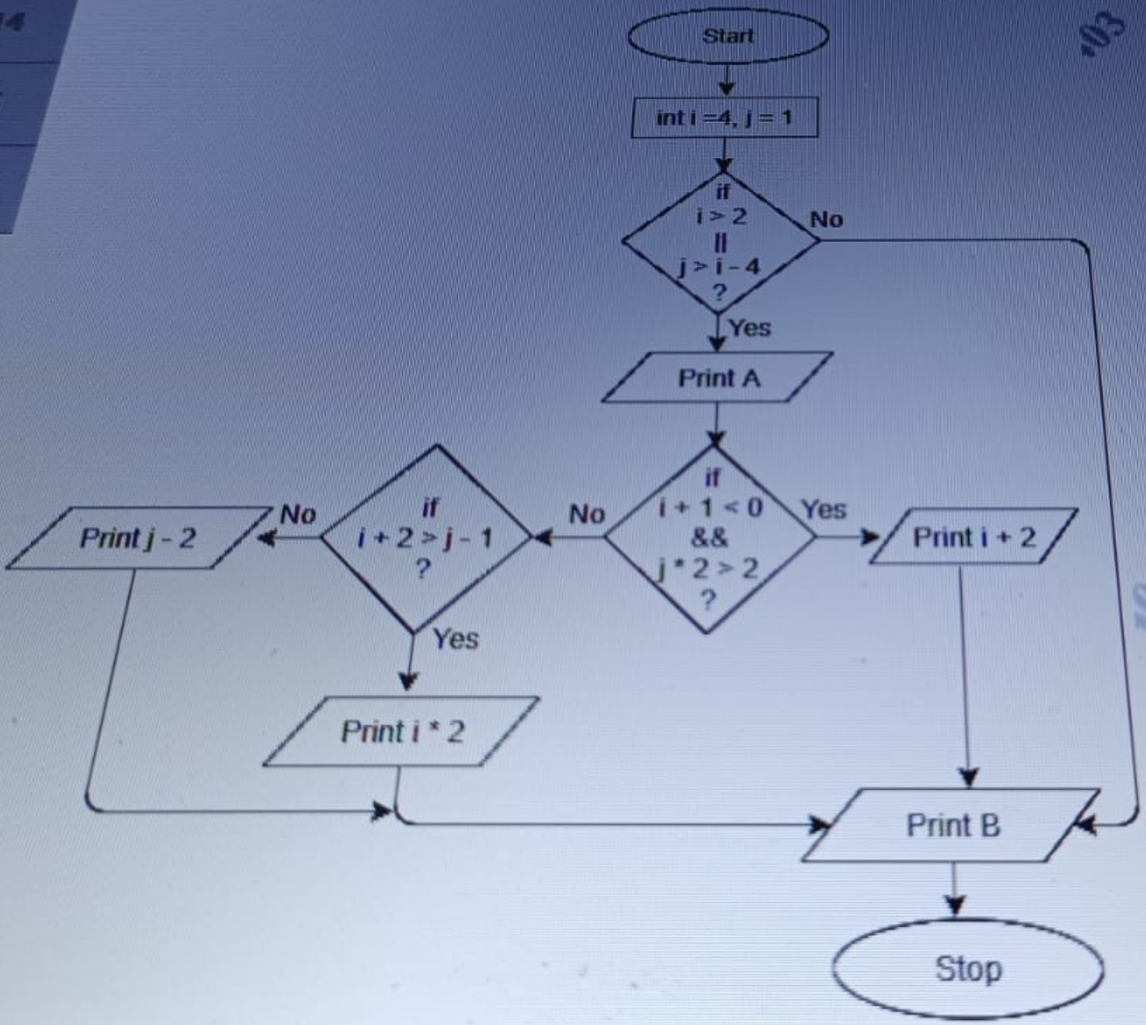
QUESTION: 17 GROUP: Technical SECTION: Programming Ability Mark(s): 2

What will be the output of the flowchart given below?



QUESTION: **18** GROUP: **Technical** SECTION: **Operating Systems** Mark(s): 2

Which one of the given options is correct for the below given procedures?

```
int temp;
int ab()
{
    return 7;
}

void abc()
{
    temp = ab();
    printf("%d", temp);
}
```



- ○ Neither procedure ab() nor abc() is re-entrant.
- ○ Procedures ab() and abc() are re-entrant.

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16 ☆
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20
21

Creational design patterns can be divided into:

I) class-creation patterns
II) object-creational patterns
III) methods-creational patterns

- Both (I) and (II)
- Only II
- Only (II) and (III)
- Only (I) and (III)

HirePro

00:30:36

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QUESTION: 20 GROUP: **Technical** SECTION: **Networking** Mark(s): 2

Programmer is working on a program to convert data into an unreadable format. The program should also convert back the data into a readable format whenever required. Which of the below methods should the programmer use?

- Encryption and Decryption
- Hashing
- Risk Analysis
- Digital Signature



Previous Question Next Question

QUESTION: **19** GROUP: **Technical** SECTION: **RDBMS and SQL** Mark(s): 2

*Which of the given below index is used to reorder the physical order of the table and search based on the key values?*

- ○ NonClustered Index
- ○ Unique Index
- ○ Function based index
- ○ Clustered Index

HirePro

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22 ☆
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QUESTION: 22 GROUP: **Technical** SECTION: **OOPS & Solid Design Principles** Mark(s): 2

Which one of the given project design violates the "Single Responsibility Principle"?

- ○ Having all the interfaces with a unique purpose.
- ○ Having the most commonly used functionalities as a base class for all the classes of project
- ○ Having multiple classes taking up exact same responsibility
- ○ Having multiple methods in the same class

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21 ☆
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QUESTION: 21 GROUP: **Technical** SECTION: **RDBMS and SQL** Mark(s): 2

Consider the table given below:

| STUDENT | | | | |
|---|---|---|---|---|
| STD_ID | STD_NAME | DATE_OF_BIRTH | PHONE | DEPT_NAME |
| 1 | aa | 10-10-1991 | 9988776655 | maths |
| 2 | bb | 11-09-1990 | 9977776655 | computers |
| 3 | cc | 02-09-1991 | 9977576655 | stats |
| 4 | dd | 02-09-1991 | 9977574655 | maths |

Which one of the query given below will display the names of students in the departments other than Maths and Computer?

- SELECT std_name FROM student WHERE dept_name NOT EXISTS ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT EXISTS IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name IN ('maths','computers');
- SELECT std_name FROM student WHERE dept_name NOT IN ('maths','computers');



Previous Question Next Question

QUESTION: **24** GROUP: **Technical** SECTION: **RDBMS and SQL** Mark(s): 2

Consider the tables given below:

| STUDENTS | | |
|---|---|---|
| **ROLLNO** | **NAME** | **PIN** |
| 5780 | kritika | 761005 |
| 5733 | Prakash | 560037 |
| 5767 | Ranjan | 500038 |
| 7739 | Nabeen | 560037 |
| 5789 | Sanjay | 560037 |
| 5797 | Janifer | 751003 |
| 5763 | Resma | 500038 |
| 5776 | Jeebitesh | 751003 |

| CITY | | |
|---|---|---|
| **PIN** | **NAME** | **POPULATION** |
| 761005 | Brahmapur | 355827 |
| 560037 | Bengalore | 11556907 |
| 500038 | Hyderabad | 11723548 |
| 751003 | Bhubaneswar | 886397 |



Which of the query/queries given below will retrieve the students whose city's population is more than 11lakhs?

```
I) SELECT s.name FROM students s, city c WHERE c.pin=s.pin and c.population>1100000;
II) SELECT name FROM students WHERE pin IN (SELECT pin FROM city WHERE population>1100000);
III) SELECT s.name FROM students s right outer join city c WHERE c.pin=s.pin and c.population>1100000;
```

○ Only (II)

ALL

○ Only (I) and (II)

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QUESTION: 25 GROUP: **Technical** SECTION: **OOPS & Solid Design Principles** Mark(s): 2

00:29:49

Assume the employee of a company has designation: Project Manager, Team Leader, and Programmer. The company wants to store information about employee such as employee ID, designation, name, age and salary. Hence a set of classes are created in Java to represent such information. Which of the following task should be performed?

- To create an Employee class and derive subclasses of Employee as Project Manager, Team Leader and Programmer.
- To create an Employee class and add fields as Project Manager, Team Leader, Programmer, employee ID, designation, name, age and salary.
- To create an Employee class and add fields as employee ID, designation, name, age and salary.
- To create a separate classes for Project Manager, Team Leader and Programmer.



Previous Question Next Question

DELL

HirePro

QUESTION: 23 GROUP: **Technical** SECTION: **Operating Systems** Mark(s): 2

Which one of the given options is correct for the below given operating systems?

I) Windows 10

II) Cent OS

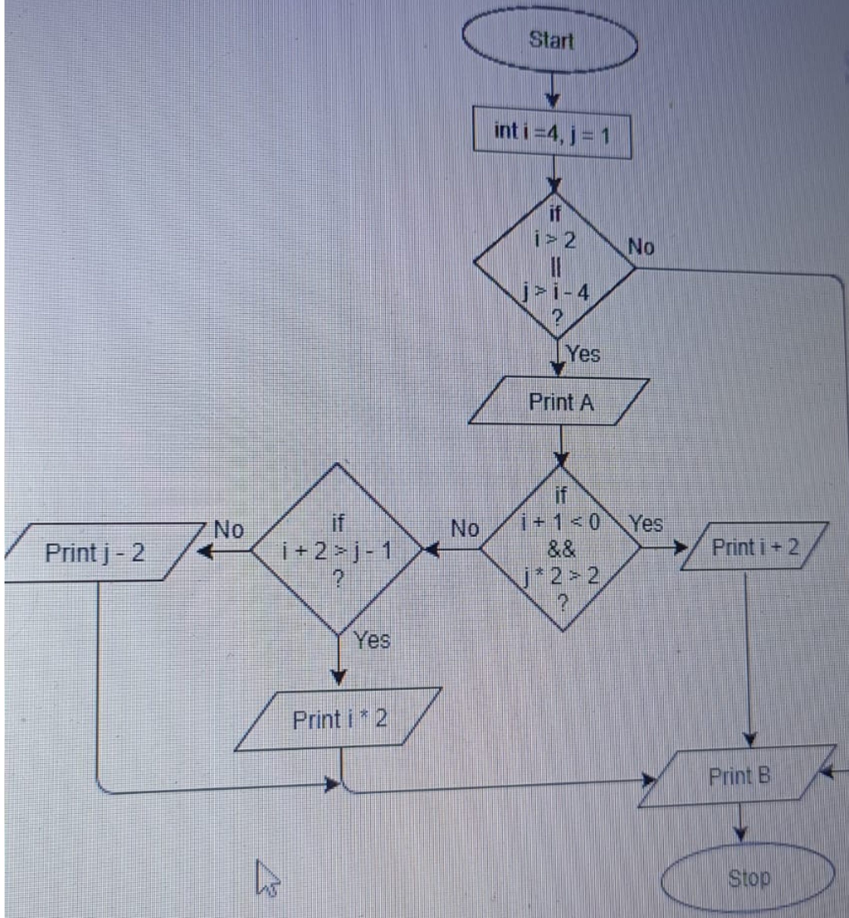
III) Gentoo

- Only (I) is a Linux distribution.
- Only (II) and (III) are Linux distributions.
- All (I), (II) and (III) are Linux distributions.
- Only (I) and (II) are Linux distributions.

14 15 16 17 18 19 20 21 22 23 24 25

QUESTION: **17** GROUP: **Technical** SECTION: **Programming Ability** Ma

What will be the output of the flowchart given below?



QUESTION 21 GROUP Technical SECTION RDBMS and SQL Mark(s) 2

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21 ☆

Consider the table given below:

| STUDENT | | | | |
|---|---|---|---|---|
| STD_ID | STD_NAME | DATE_OF_BIRTH | PHONE | DEPT_NAME |
| 1 | aa | 10-10-1991 | 9988776655 | maths |
| 2 | bb | 11-09-1990 | 9977776655 | computers |
| 3 | cc | 02-09-1991 | 9977576655 | stats |
| 4 | dd | 02-09-1991 | 9977574655 | maths |



Which one of the query given below will display the names of students in the departments other than Maths and Computer?

- ○ SELECT std_name FROM student WHERE dept_name NOT EXISTS ('maths','computers');
- ○ SELECT std_name FROM student WHERE dept_name NOT EXISTS IN ('maths','computers');
- ○ SELECT std_name FROM student WHERE dept_name IN ('maths','computers');
- ◉ SELECT std_name FROM student WHERE dept_name NOT IN ('maths','computers');

Unanswer

14

15 ☆

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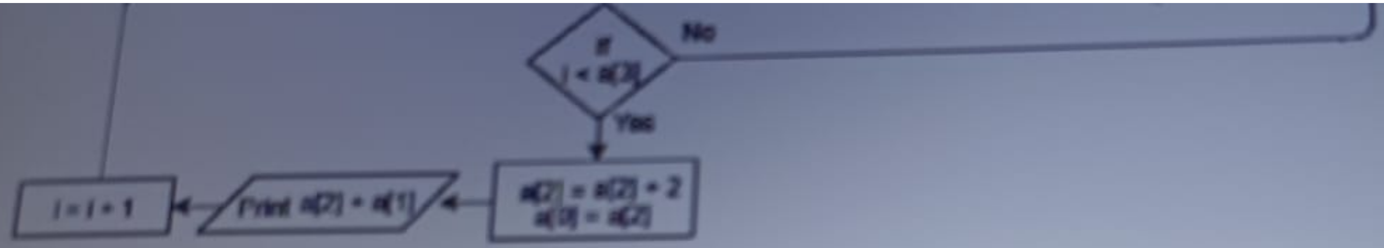


- 5 1 3 4 4
- 5 2 9 1 2
- 6 1 8 2 1
- 4 2 3 4 7

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QUESTION 27 GROUP Coding SECTION Coding 1 Mark(s) 20

27 ☆ A group of users on a network want access to a printer for printing a certain number of pages T. This network is connected to a group of printers. These printers have limited ink and due to this, the printers have been set to print only t pages. Here, t pages will be calculated as the total number of factors of the number of pages T specified by the user.

The number of printers that are available (N) and the number of pages (T) is specified by the user.
Zero-indexed array means an array whose index starts from 0. Hence, the first element's index is 0; the second element's index is 1, the kth element's index is (k-1) and so on.

Write a program to determine the number of pages that can be printed from each printer A[i] such that 0 ≤ i < N.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

**Constraints:**
I) 1 <= N <= 50000
II) 1 < A[i] < 100000

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**Input Format:**
The first line contains the number of printers N.
The next N lines contain the total number of pages T to be printed that are given by the user.

**Output Format:**
N lines of output contain t, number of pages selected for printing for each printer.

**Sample Input 1:**
5
4
5
2
3
6

**Sample Output 1:**



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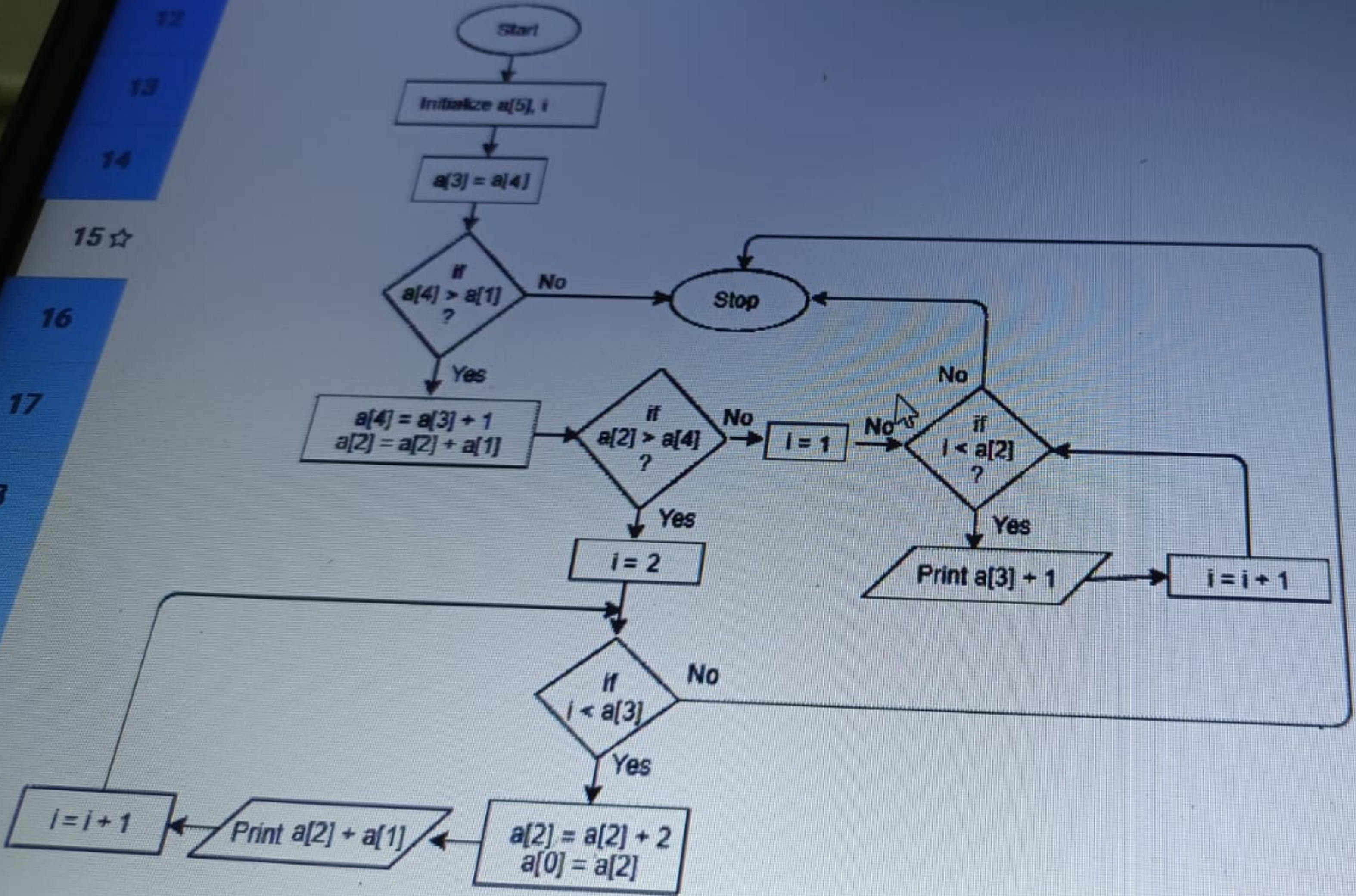
QUESTION 15 GROUP Technical SECTION Programming Ability Mark(s): 2

Which set of array elements given in options will give the output as "15 17 19 21 23 25" if passed to the flowchart given below?



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27 ☆

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**Explanation 2:**

Total number of printers available are 3.
For the first printer, T=12. The number of factors of integer '12' are 6 (1,2,3,4,6 and 12), which is printed in the first line of the output.
For the second printer, T=22. the number of factors of integer '22' are 4 (1,2,11 and 22), which is printed in the second line of the output.
For the second printer, T=9. the number of factors of integer '9' are 3 (1,3 and 9), which is printed in the third line of the output.

Current Code | Previous Versions | C++

```
#include <bits/stdc++.h>
using namespace std;

int totalDivisors(int N,int array[]) // N contains the total number of printers.array is an array of N pages to be printed
{
    int count = 0;

    // WRITE YOUR CODE HERE

    return count;
}

int main() {

    int N;
    cin >> N;

    int array[N];
    for(int i=0; i<N; i++)
        cin >> array[i];

    totalDivisors(N,array);
}
```

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27 ☆

3
6

**Sample Output 1:**
3
2
2
2
4

**Explanation 1:**
The total number of printers available are 5.
For the first printer, T=4. The number of factors of integer '4' are 3 (1,2 and 4), which is printed in the first line of the output.
For the second printer, T=5. The number of factors of integer '5' are 2 (1 and 5), which is printed in the second line of the output.
Similarly, the number of pages that can be printed is found out for the following printers.

**Sample Input 1:**
3
12
22
9

**Sample Output 1:**
6
4
3

**Explanation 2:**
Total number of printers available are 3.
For the first printer, T=12. The number of factors of integer '12' are 6 (1,2,3,4,6 and 12), which is printed in the first line of the output.
For the second printer, T=22. the number of factors of integer '22' are 4 (1,2,11 and 22), which is printed in the second line of the output.
For the second printer, T=9. the number of factors of integer '9' are 3 (1,3 and 9), which is printed in the third line of the output.

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